



**POLITECNICO
MILANO 1863**

**COMPUTER SCIENCE AND ENGINEERING
INTERNET OF THINGS (IoT)
2025 - 2026**

Homework RFID

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Exercise 3 - RFID

A RFID system based on Dynamic Frame ALOHA is composed of $N=4$ tags.

EQ1:

Find the overall collision resolution efficiency η in the different cases in which the initial frame size is set to $r_1 = 1, 2, 3, 4, 5, 6$.

- Assume that after the first frame, the frame size is correctly set to the current backlog size.
- Assume as given the duration of the arbitration period with $N = 2, 3$ tags when $r = N$ ($L_2 = 4, L_3 = \frac{51}{8}$).

The overall collision resolution efficiency is given by $\eta = \frac{N}{L_N^*}$, with $N = 4$ and

$$\begin{aligned} L_N^* &= r_1 + \sum_{i=0}^{N-1} P(S = i) \cdot L_{N-i} \\ &= r_1 + P(S = 0) \cdot L_4 + P(S = 1) \cdot L_3 + P(S = 2) \cdot L_2 + P(S = 3) \cdot L_1 \end{aligned}$$

Therefore, we should calculate L_4^* for different initial frame sizes, $r_1 = 1, 2, 3, 4, 5, 6$, and then determine the corresponding efficiency.

Based on the assumption, after the first frame, the frame size is set equal to the current backlog size. We also know the arbitration period of $L_1 = 1, L_2 = 4$ and $L_3 = \frac{51}{8}$.

Therefore we should calculate the arbitration period of L_4 .

$$\begin{aligned} L_4 &= r + \sum_{i=0}^{N-1} P(S = i) \cdot L_{N-i} \\ &= r + P(S = 0) \cdot L_4 + P(S = 1) \cdot L_3 + P(S = 2) \cdot L_2 + P(S = 3) \cdot L_1 \end{aligned}$$

where:

- $N = 4$ tags to be resolved with $r = N = 4$ slots.
- Probability that none of the 4 tags is resolved in 4 slots:

$$P(S = 0) = \left(\frac{1}{4}\right)^4 \cdot \left(4 + \binom{4}{2} \cdot \binom{4}{2}\right) = \frac{5}{32}$$

- Probability that 1 of the 4 tags is resolved in 4 slots:

$$P(S = 1) = \left(\frac{1}{4}\right)^4 \cdot 4 \cdot 4 \cdot 3 = \frac{3}{16}$$

- Probability that 2 of the 4 tags are resolved in 4 slots:

$$P(S = 2) = \left(\frac{1}{4}\right)^4 \cdot 4 \cdot 3 \cdot \binom{4}{2} \cdot 2 = \frac{9}{16}$$

- Probability that 3 of the 4 tags are resolved in 4 slots:

$$P(S = 3) = 0$$

- Probability that all of the 4 tags is resolved in 4 slots:

$$P(S = 4) = \left(\frac{1}{4}\right)^4 \cdot 4! = \frac{3}{32}$$

therefore, $L_4 = 4 + \frac{5}{32} \cdot L_4 + \frac{3}{16} \cdot \frac{51}{8} + \frac{9}{16} \cdot 4 + 0 \cdot 1 \Rightarrow L_4 = \frac{953}{108}$.

Case 1: there are $N = 4$ tags to be resolved using an initial frame of 1 slot.

$$L_4^* = 1 + P(S = 0) \cdot \frac{953}{108} + P(S = 1) \cdot \frac{51}{8} + P(S = 2) \cdot 4 + P(S = 3) \cdot 1$$

where:

- Probability that none of the 4 tags is resolved in 1 slot:

$$P(S = 0) = 1$$

- Probability that 1/2/3/4 of the 4 tags are resolved in 1 slot:

$$P(S = 1) = P(S = 2) = P(S = 3) = P(S = 4) = 0$$

therefore, $L_4^* = 1 + 1 \cdot \frac{953}{108} + 0 \cdot \frac{51}{8} + 0 \cdot 4 + 0 \cdot 1 = 9.824074074$ and

$$\eta_1 = \frac{4}{9.824074074} \simeq 0.4072.$$

Case 2: there are $N = 4$ tags to be resolved using an initial frame of 2 slots.

$$L_4^* = 2 + P(S = 0) \cdot \frac{953}{108} + P(S = 1) \cdot \frac{51}{8} + P(S = 2) \cdot 4 + P(S = 3) \cdot 1$$

where:

- Probability that none of the 4 tags is resolved in 2 slots:

$$P(S = 0) = \left(\frac{1}{2}\right)^4 \cdot (2 + \binom{4}{2} \cdot \binom{2}{2}) = \frac{1}{2}$$

- Probability that 1 of the 4 tags is resolved in 2 slots:

$$P(S = 1) = \left(\frac{1}{2}\right)^4 \cdot 4 \cdot 2 = \frac{1}{2}$$

- Probability that 2/3/4 of the 4 tags are resolved in 2 slots:

$$P(S = 2) = P(S = 3) = P(S = 4) = 0$$

therefore, $L_4^* = 2 + \frac{1}{2} \cdot \frac{953}{108} + \frac{1}{2} \cdot \frac{51}{8} + 0 \cdot 4 + 0 \cdot 1 = 9.599537037$ and

$$\eta_2 = \frac{4}{9.599537037} \simeq 0.4167.$$

Case 3: there are $N = 4$ tags to be resolved using an initial frame of 3 slots.

$$L_4^* = 3 + P(S = 0) \cdot \frac{953}{108} + P(S = 1) \cdot \frac{51}{8} + P(S = 2) \cdot 4 + P(S = 3) \cdot 1$$

where:

- Probability that none of the 4 tags is resolved in 3 slots:

$$P(S = 0) = \left(\frac{1}{3}\right)^4 \cdot (3 + \binom{4}{2} \cdot \binom{3}{2}) = \frac{7}{27}$$

- Probability that 1 of the 4 tags is resolved in 3 slots:

$$P(S = 1) = \left(\frac{1}{3}\right)^4 \cdot 4 \cdot 3 \cdot 2 = \frac{8}{27}$$

- Probability that 2 of the 4 tags are resolved in 3 slots:

$$P(S = 2) = \left(\frac{1}{3}\right)^4 \cdot \binom{4}{2} \cdot 3 \cdot 2 = \frac{4}{9}$$

- Probability that 3/4 of the 4 tags are resolved in 3 slots:

$$P(S = 3) = P(S = 4) = 0$$

therefore, $L_4^* = 3 + \frac{7}{27} \cdot \frac{953}{108} + \frac{8}{27} \cdot \frac{51}{8} + \frac{4}{9} \cdot 4 + 0 \cdot 1 = 8.954389575$ and

$$\eta_3 = \frac{4}{8.954389575} \simeq 0.4467.$$

Case 4: there are $N = 4$ tags to be resolved using an initial frame of 4 slots.

$$L_4^* = L_4 = 8.824074074 \text{ and}$$

$$\eta_4 = \frac{4}{8.824074074} \simeq 0.4533.$$

Case 5: there are $N = 4$ tags to be resolved using an initial frame of 5 slots.

$$L_4^* = 5 + P(S = 0) \cdot \frac{953}{108} + P(S = 1) \cdot \frac{51}{8} + P(S = 2) \cdot 4 + P(S = 3) \cdot 1$$

where:

- Probability that none of the 4 tags is resolved in 5 slots:

$$P(S = 0) = \left(\frac{1}{5}\right)^4 \cdot (5 + \binom{4}{2} \cdot \binom{5}{2}) = \frac{13}{125}$$

- Probability that 1 of the 4 tags is resolved in 5 slots:

$$P(S = 1) = \left(\frac{1}{5}\right)^4 \cdot 4 \cdot 5 \cdot 4 = \frac{16}{125}$$

- Probability that 2 of the 4 tags are resolved in 5 slots:

$$P(S = 2) = \left(\frac{1}{5}\right)^4 \cdot \binom{4}{2} \cdot 5 \cdot 4 \cdot 3 = \frac{72}{125}$$

- Probability that 3 of the 4 tags are resolved in 5 slots:

$$P(S = 3) = 0$$

- Probability that all of the 4 tags is resolved in 5 slots:

$$P(S = 4) = \left(\frac{1}{5}\right)^4 \cdot 4! \cdot 5 = \frac{24}{125}$$

$$\text{therefore, } L_4^* = 5 + \frac{13}{125} \cdot \frac{953}{108} + \frac{16}{125} \cdot \frac{51}{8} + \frac{72}{125} \cdot 4 + 0 \cdot 1 = 9.037703704 \text{ and}$$

$$\eta_5 = \frac{4}{9.037703704} \simeq 0.4426.$$

Case 6: there are $N = 4$ tags to be resolved using an initial frame of 6 slots.

$$L_4^* = 6 + P(S = 0) \cdot \frac{953}{108} + P(S = 1) \cdot \frac{51}{8} + P(S = 2) \cdot 4 + P(S = 3) \cdot 1$$

where:

- Probability that none of the 4 tags is resolved in 6 slots:

$$P(S = 0) = \left(\frac{1}{6}\right)^4 \cdot (6 + \binom{4}{2} \cdot \binom{6}{2}) = \frac{2}{27}$$

- Probability that 1 of the 4 tags is resolved in 6 slots:

$$P(S = 1) = \left(\frac{1}{6}\right)^4 \cdot 4 \cdot 6 \cdot 5 = \frac{5}{54}$$

- Probability that 2 of the 4 tags are resolved in 6 slots:

$$P(S = 2) = \left(\frac{1}{6}\right)^4 \cdot \binom{4}{2} \cdot 6 \cdot 5 \cdot 4 = \frac{5}{9}$$

- Probability that 3 of the 4 tags are resolved in 6 slots:

$$P(S = 3) = 0$$

- Probability that all of the 4 tags is resolved in 6 slots:

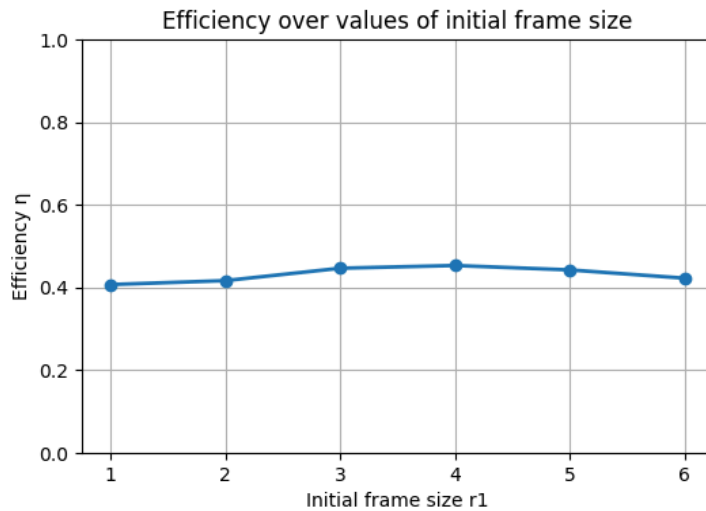
$$P(S = 4) = \left(\frac{1}{6}\right)^4 \cdot 4! \cdot \binom{6}{4} = \frac{5}{18}$$

$$\text{therefore, } L_4^* = 6 + \frac{2}{27} \cdot \frac{953}{108} + \frac{5}{54} \cdot \frac{51}{8} + \frac{5}{9} \cdot 4 + 0 \cdot 1 = 9.466135117 \text{ and}$$

$$\eta_6 = \frac{4}{9.466135117} \simeq 0.4226.$$

EQ2:

After computing the values of the efficiency with the different frame sizes, produce a plot with values of η over r_1 .



r_1	Efficiency η
1	0.4072
2	0.4167
3	0.4467
4	0.4533
5	0.4426
6	0.4226

EQ3:

For what values of r_1 we have the maximum value for η ? Briefly comment.

For $r_1 = 4$, the efficiency reaches its maximum value, with $\eta = 0.4533$. This is expected because, in this case, the initial frame size matches the number of tags, so $r_1 = N = 4$.

When the frame size is smaller than the number of tags, more tags collide in the same slots; on the other hand, when the frame size is too large, many slots remain empty. Therefore $r_1 = 4$ provides the best balance in Dynamic Frame ALOHA.